

Duc Duy Nguyen

ducduynguyencs@gmail.com | (+84) 357735494 | Cau Giay, Ha Noi, Viet Nam

EDUCATION

Posts and Telecommunications Institute of Technology

Aug. 2018 – Mar. 2023

B.Eng in Information Technology.

Thesis: Remote monitoring of human vital signs using mm-wave FMCW radar.

RESEARCH AND WORK EXPERIENCES

VinAI

Mar. 2023 – Jan. 2025

AI Engineer

- Designed and conducted quality control for a controlled study investigating the impact of alcohol on vehicle operation, collecting data from multimodal sensors such as vehicle information systems, alcohol sensors, and infrared cameras.
- Performed data analysis and feature engineering to identify how intoxicated behavior manifests across different modalities.
- Developed a novel system capable of passively detecting intoxicated drivers through multimodal fusion, which was patented in the U.S. and successfully demonstrated at the InCabin event in Europe.
- Focused on developing innovative methods for tracking and recognizing human actions using mmWave Frequency-Modulated Continuous-Wave (FMCW) radar under occlusion scenarios. This project was funded by the Air Force Office of Scientific Research (AFOSR).

VinAI

Mar. 2022 – Mar. 2023

Contract AI Engineer

- Designed and conducted quality control for a study collecting drivers' respiration and heart rates while driving.
- Leveraged an encoder-decoder architecture to remotely and passively monitor drivers' respiration and heart rates using mmWave radar signals.
- Explored the potential of thermal sensors for detecting intoxicated drivers.
- Proposed novel state-of-the-art class-agnostic methods in a few-shot setting to count human actions of interest using inertial sensor data from wearable devices. This work resulted in a submission at TPAMI and a publication at AAAI.

Naver AI Laboratory

Jun. 2020 – Mar. 2022

Research Assistant

- Collected, cleaned, and preprocessed inertial sensor data for analysis.
- Conducted literature reviews and implemented state-of-the-art methods for human activity recognition and repetitive action counting using sensor data.
- Developed Android applications for smartphones and wearable devices to recognize and track human activities in real-time.

PUBLICATIONS AND PATENTS

1. **Duc Duy Nguyen**, Yifeng Huang, Lam Nguyen, Cuong Pham and Minh Hoai. Count What You Want: Exemplar Identification and Few-Shot Counting of Human Actions in the Wild. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2024. Doi: 10.1609/aaai.v38i9.28869.
2. **Duc Duy Nguyen**, Lam Nguyen, Yifeng Huang, Cuong Pham, Minh Hoai. Class-Agnostic Repetitive Action Counting for Wearable Devices. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, Under Review, 2024.
3. **Duc Duy Nguyen**, De Vu, Lam Nguyen, Tien Tran, Hung Bui. Multimodal Sensor and Vision-based Driver Alcohol Impairment Detection System. Filing US Patent No.: 18/642,354.

HONOURS AND AWARDS

- Innovation Award Honoree by Consumer Technology Association (CES) for VinFast Mirrorsense (2024)
- Award for outstanding academic performance by Posts and Telecommunications Institute of Technology (2023)
- Top 5 AI Awards by AI4VN for VinAI Driver and Occupancy Monitoring System (2022)
- Scholarship for excellent performance by Posts and Telecommunications Institute of Technology (2018-2020)
- 3rd place in the Physics Olympiad competition organized by Le Quy Don High School, Ha Noi, Viet Nam (2016)
- 1st place in the Physics Olympiad competition organized by Le Quy Don High School, Ha Noi, Viet Nam (2015)